

CODE SPORT



In this month's column, we discuss foundation large language models and how they are being claimed to be the foundation for broader AI.

The last couple of years have seen a number of real world impactful developments in AI, driven primarily by a single large class of deep learning models. These models have their origin in the 'transformer' model developed in 2017, in the seminal paper 'Attention is all you need' by Vaswani *et al.* In the NLP space, almost all of the State of the Art (SoTA) models for benchmarking have been based on transformer variants such as BERT, Roberta, GPT-2, etc. We have also discussed a number of these models in our earlier columns.

Of late, transformer models have also been adopted for computer vision with Vision Transformers (ViT), beating standard CV models in many of the popular models. Multi-modal transformer models have also been developed for combined language and vision tasks such as visual question answering, text based image retrieval, visual common sense reasoning, etc. All these SoTA transformer models have been trained on very large data, and their performance capabilities have been attributed both to their 'transformer' based architecture and the vast quantities of data they have been trained on. Recent research in the AI space has defined this particular class of models as foundation models, which means that these models will be the foundation for deeper/broader AI systems of the future. In this column, we will discuss 'foundation models', examining their capabilities and shortcomings, and also opine on whether they are really foundational to AI.

The story of foundation models starts with large language models. A language model is a deep learning model trained to predict the next word in a text sequence. For example, consider a sentence 'the quick brown fox jumped over the fence', where the language model has so far seen the sentence 'the

quick brown fox' and is asked to predict the next word. Let us say that it predicts the next word as 'ran' instead of 'jumped', as present in the ground reality. The model will take this error, back propagate it across the network, and adjust the weights so that the probability for completion for the next word being 'jumped' is higher compared to the word 'ran'. Of course, as in all deep learning models, word tokens are represented as embedding vectors and at the output, we will have a Softmax layer. The result of this is the probabilities of all possible output words. The word that has the maximum Softmax probability among the output vocabulary words will be taken as the next predicted word.

You may be wondering how can this model, which is trained on the 'next word prediction' task, be useful for various other NLP tasks, which are not about 'next word prediction'. For instance, how can we use this model for a task such as sentiment analysis or text classification? This is pretty straight forward — we take the language model, remove the Softmax layer for the next word prediction, and put a simple classifier network on top of it. The classifier network can be as simple as a single feed forward layer followed by a Softmax layer. Basically, the language model serves as the backbone for any other NLP task you want to solve. And we then fine-tune the task-specific data, which is typically of much smaller size. Hence, we can take the same pretrained language model and apply it to multiple different tasks by fine-tuning it individually on each of the task-specific data sets. This is the principle of transfer learning.

You may be wondering why the model pretrained on the 'next word prediction' task can perform well even if you fine-tune it on a small amount of task-specific data. It has been shown that a number of general capabilities/features needed for